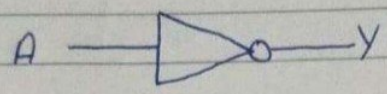
BASIC LOGIC GATES

Basic Logic Gates

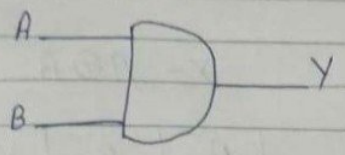
1. NOT Gate



$Y = \bar{A}$

A	Y
0	1
1	0

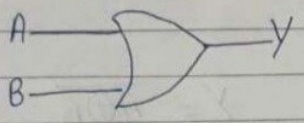
2. AND Gate



$Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

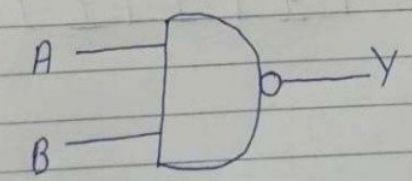
3. OR Gate



$Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

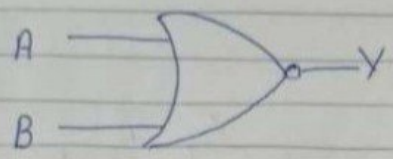
4. NAND Gate



$Y = \overline{A \cdot B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5. NOR Gate

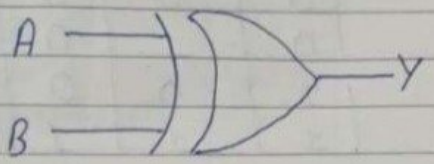


$Y = \overline{A+B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6. XOR Gate

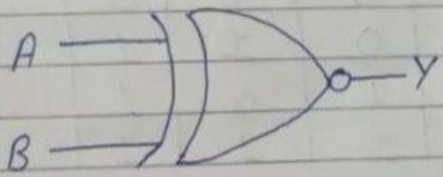
$Y = A \oplus B$



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

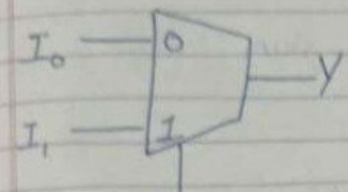
7. XNOR Gate

$Y = \overline{A \oplus B}$



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

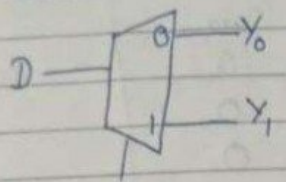
8. MUX(2:1)



S	Y
0	I_0
1	I_1

S MUX $Y = I_0\bar{S} + I_1 S$

9. DMUX (1:2)

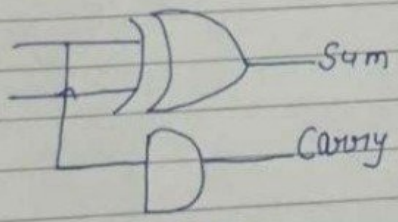


S	Y_0	Y_1
0	D	0
1	0	D

S DMUX $Y_0 = D\bar{S}$

$Y_1 = DS$

10 Half Adder

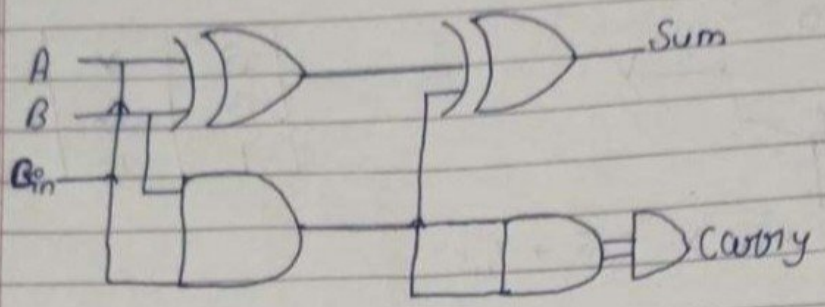


$Sum = A \oplus B$

$Carry = A \cdot B$

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

II. Full Adder

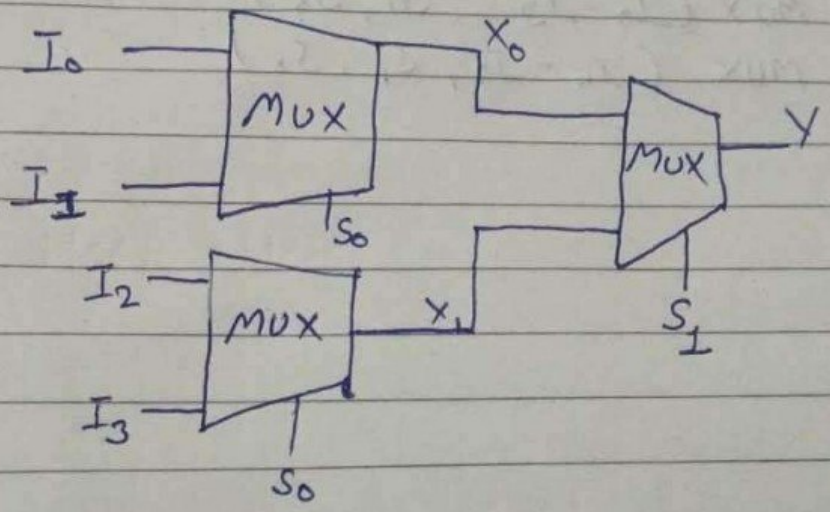


$Sum = A \oplus B \oplus C_{in}$

$Carry = A \cdot B + C_{in}(A \oplus B)$

A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

MUX (4:1)

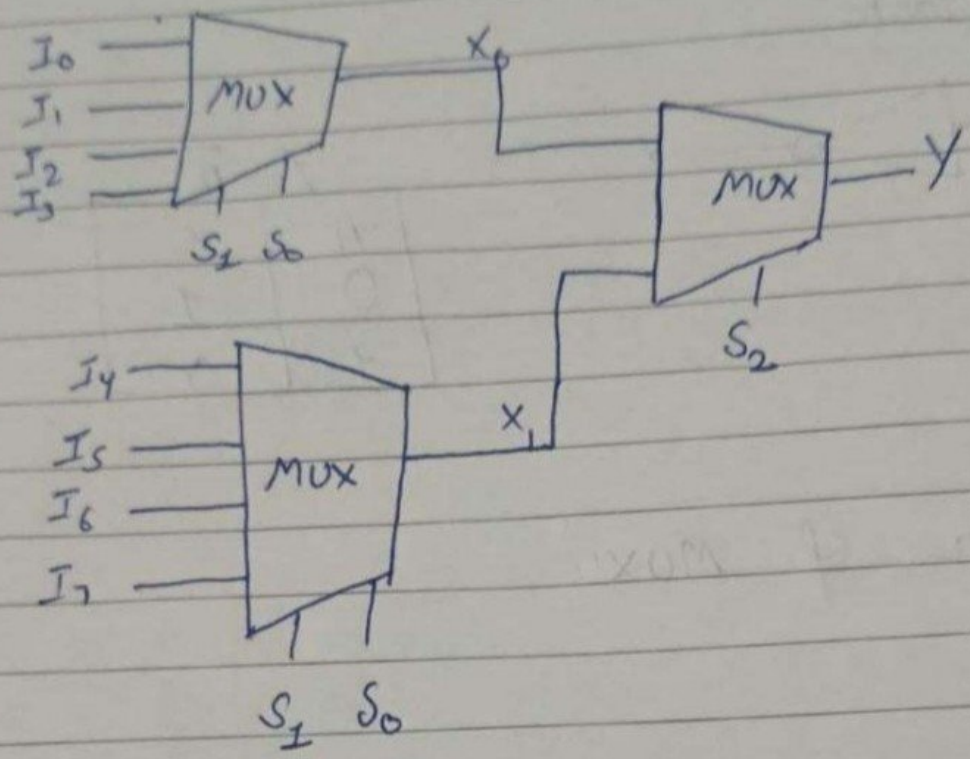


$Y = MUX(X_0, X_1, S_1)$

where $X_0 = MUX(\bar{I}_0, I_1, S_0)$

$X_1 = MUX(I_2, I_3, S_0)$

MUX (8:1)

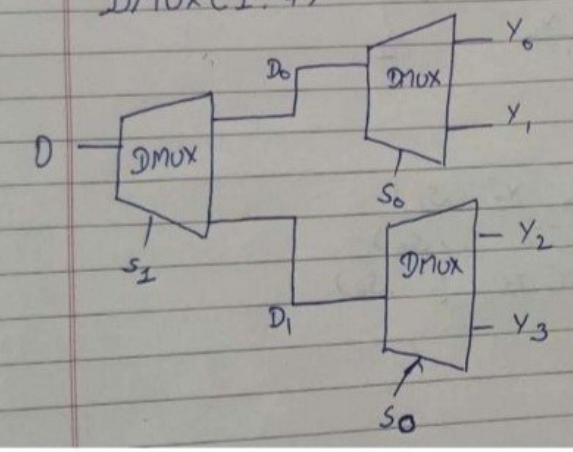


$Y = MUX(X_0, X_1, S_2)$

where $X_0 = MUX(I_0 - I_3, S_1, S_0)$

$X_1 = MUX(I_4 - I_7, S_1, S_0)$

DMUX(1:4)



$Y_0, Y_1 = DMux(D_0, S_0)$
$Y_2, Y_3 = DMux(D_1, S_0)$
where $D_0, D_1 = DMux(D, S_1)$

DMUX (1:8)

